



## Project Overview

---

AirBnB is a global vacation rental platform that connects property owners (hosts) with travellers seeking accommodations. The platform features millions of listings, from budget-friendly apartments to luxurious villas. To remain competitive, AirBnB continuously analyses its data to improve user experiences, optimise pricing, and enhance host and guest satisfaction.

The analytics team at AirBnB is tasked with delivering insights into key performance areas, such as host activity, booking trends, and guest feedback. By leveraging the platform's data, they aim to optimise revenue, improve guest satisfaction, and identify areas of growth or underperformance.



## Goal

---

The goal is to extract actionable insights from AirBnB's datasets, focusing on:

- ☐ Booking trends and occupancy rates.
- ☐ Host performance and listing popularity.
- ☐ Guest feedback and property diversity.



## Dataset Description & Schema

---

AirBnB\_Analysis/AirBnB Dataset.csv at main · shivangagrahari02-web/AirBnB\_Analysis

### Listings Table

Tracks detailed information about properties listed on AirBnB:

- **id**: Unique identifier for the listing.
- **name**: Title of the listing.
- **description**: Detailed property description.
- **host\_id**: Identifier for the host managing the property.
- **host\_name**: Name of the host.
- **property\_type**: Type of property (e.g., house, apartment).
- **room\_type**: Type of room (e.g., entire home, private room).
- **accommodates**: Maximum number of guests.
- **price**: Price per night.
- **review\_scores\_rating**: Average guest rating.
- **number\_of\_reviews**: Total number of reviews.

## Calendar Table

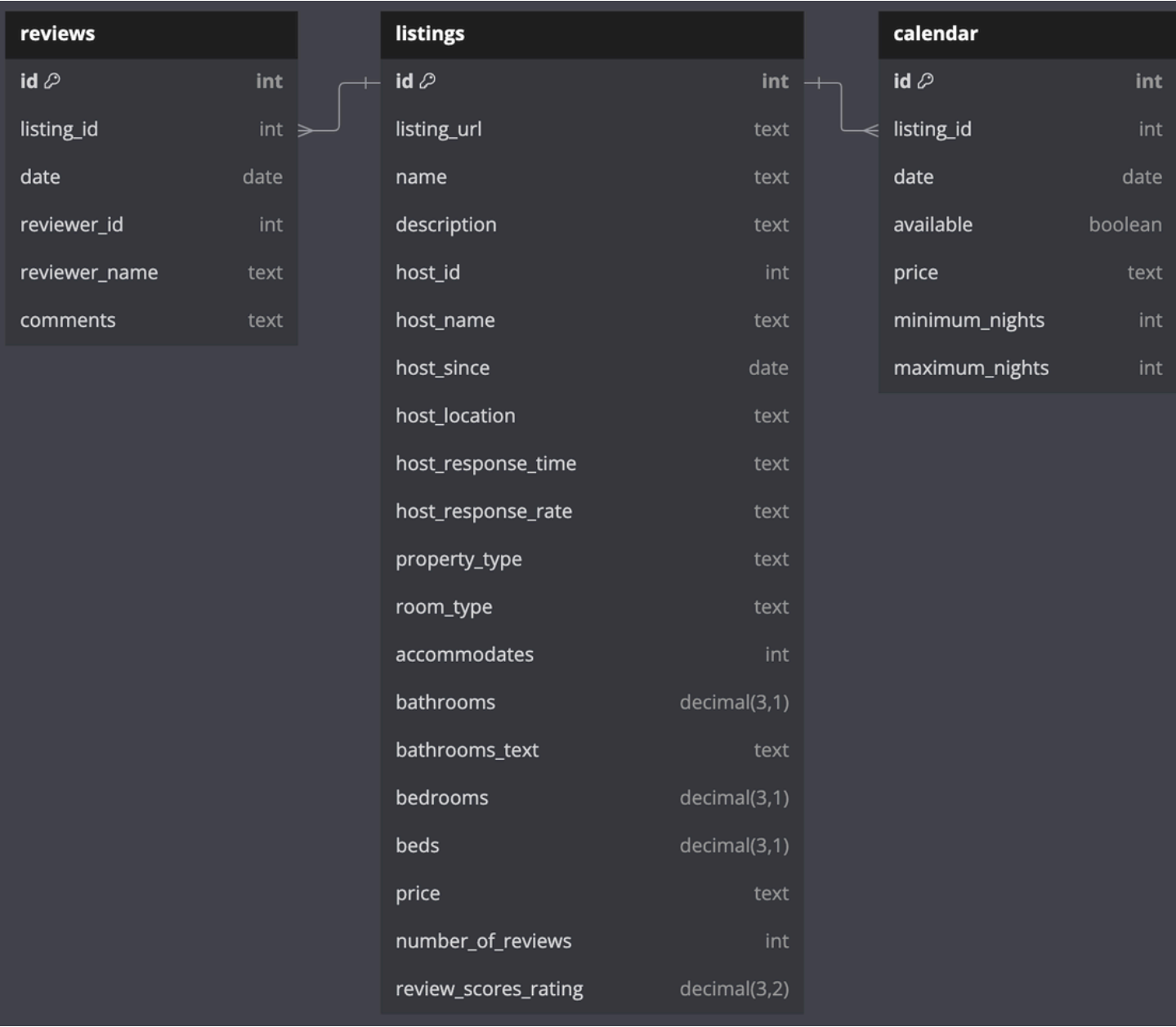
Captures booking and availability data:

- **listing\_id**: Foreign key linking to the id column in the listings table.
- **date**: Specific date for the entry.
- **available**: Indicates if the listing is available (t for true, f for false).
- **price**: Nightly price for the listing.

## Reviews Table

Records guest feedback:

- **listing\_id**: Foreign key linking to the id column in the listings table.
- **id**: Unique identifier for each review.
- **date**: Date the review was posted.
- **reviewer\_id**: Identifier for the reviewer.
- **reviewer\_name**: Name of the reviewer.
- **comments**: Text of the guest's feedback.



# Analysis & Queries

# shivangagrahari02-web/ **AirBnB\_Analysis**



Contributor



Issues



Stars



Forks



## Property Diversity

With a wide range of property types available, from apartments and villas to unique stays like treehouses and boats, the management team wants to evaluate whether they are meeting customer needs.

Highlighting the most common property types and uncover underserved categories.

1. The total number of listings and unique property types.



```
SELECT
```

```
  COUNT(id) AS total_number_of_listings,
```

```
  COUNT(DISTINCT property_type) AS unique_property_types
```

```
FROM listings;
```

|   | total_number_of_listing<br>bigint | unique_property_type<br>bigint |
|---|-----------------------------------|--------------------------------|
| 1 | 419                               | 21                             |

The platform offers a broad range of property types, indicating Airbnb's ability to serve multiple traveler segments ranging from budget travelers to luxury vacation seekers. However, if a small number of property types dominate inventory, Airbnb becomes vulnerable to shifts in traveler preferences. The supply acquisition team should target hosts operating in underrepresented categories such as cabins,

cottages, or unique stays and aim to increase their inventory share by 10–15% over the next year to diversify platform offerings and reduce category concentration risk.

2. The top 5 most common property types and their respective counts.

|                                                                                   |                                              |
|-----------------------------------------------------------------------------------|----------------------------------------------|
|  | SELECT                                       |
|                                                                                   | property_type, COUNT(*) AS respective_counts |
|                                                                                   | FROM listings                                |
|                                                                                   | GROUP BY property_type                       |
|                                                                                   | ORDER BY respective_counts DESC              |
|                                                                                   | LIMIT 5;                                     |

|   | property_type<br>text       | respective_counts<br>bigint |
|---|-----------------------------|-----------------------------|
| 1 | Entire rental unit          | 203                         |
| 2 | Private room in home        | 67                          |
| 3 | Entire home                 | 63                          |
| 4 | Private room in rental u... | 19                          |
| 5 | Room in hotel               | 10                          |

A significant portion of Airbnb inventory is concentrated within a few property categories, indicating where customer demand is strongest. These categories are likely responsible for a large share of bookings and revenue. The marketing team should prioritize these property types in paid acquisition campaigns while simultaneously launching host incentive programs for lower-supply categories. Success should be measured through growth in bookings for non-dominant property types and reduced dependence on the top categories.

## Guest Ratings

Guest satisfaction is critical to Airbnb’s success, as high ratings lead to repeat bookings and positive word-of-mouth. Identifying top-rated listings can help Airbnb feature these properties in promotional campaigns and inspire other hosts to improve their offerings.

Conversely, understanding listings with poor ratings will allow Airbnb to address guest concerns through targeted interventions, such as host training or quality assurance checks.

1. Calculate the average review score across all listings.

|                                                                                   |                                                                                                  |
|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
|  | <pre> SELECT      ROUND(AVG(review_scores_rating), 2) AS avg_review_score  FROM listings; </pre> |
|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|

|   | avg_review_score |
|---|------------------|
|   | numeric          |
| 1 | 4.76             |

The overall review score serves as a platform-wide customer satisfaction indicator. A high average rating suggests Airbnb is consistently meeting guest expectations, while a declining average would signal operational or service quality issues. The customer experience team should establish a minimum platform-wide rating benchmark and continuously monitor rating trends by geography and property type. A target improvement of at least 0.1 rating points annually can significantly improve booking conversion rates and guest trust.

2. List the top 10 listings with the highest review scores.

|                                                                                     |                                                                                                                                                                     |
|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre> SELECT      name, review_scores_rating  FROM listings  WHERE review_scores_rating IS NOT NULL  ORDER BY review_scores_rating DESC, name ASC  LIMIT 10; </pre> |
|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                               | name                                               | review_scores_rating |
|-----------------------------------------------|----------------------------------------------------|----------------------|
|                                               | text                                               | numeric              |
| 1                                             | Renovated 2br/2ba, close to Alb Med & St. Petes.   | 5                    |
| 2                                             | Quiet 2-bedroom condo in the heart of Albany       | 5                    |
| 3                                             | Luxury Albany Apt. Secure, Modern, w/ Parking      | 5                    |
| 4                                             | Spacious private suite and private bath            | 5                    |
| 5                                             | /zBig Blue Ranch\ 1962 SUNY Eagle Hill 4Beds 2B... | 5                    |
| 6                                             | Elegant Studio in Center Square                    | 5                    |
| Total rows: 10    Query complete 00:00:00.238 |                                                    |                      |

Top-rated listings consistently deliver exceptional guest experiences and represent operational best practices. These listings can be used as benchmarks for host education programs. Airbnb should analyze common characteristics such as pricing strategy, amenities, response time, and property presentation, then incorporate these findings into host onboarding and training initiatives. Success can be measured by increasing the percentage of listings achieving ratings above 4.8.

3. Count the number of listings with an average review score below 4.0.



```
SELECT
    name,
    ROUND(AVG(review_scores_rating)::NUMERIC, 2) AS avg_rating
FROM listings
GROUP BY name
HAVING AVG(review_scores_rating) < 4.0;
```

|                | name<br>text                                      | round<br>numeric            |
|----------------|---------------------------------------------------|-----------------------------|
| 1              | Central Albany Apartment: Walk to State Capitol!  | 3.96                        |
| 2              | Albany-3 bedrooms apt (2nd fl) A&A-Sisters Ren... | 3.25                        |
| 3              | Private bedroom close to St. peter and Albany ... | 3.67                        |
| 4              | Center Square Serenity: 2-Bedroom Elegance        | 3.50                        |
| 5              | Cozy 1 bedroom with office and luxurious King ... | 3.00                        |
| 6              | /An Oasis 8 ppl Yard Grill W/D\                   | 3.00                        |
| Total rows: 10 |                                                   | Query complete 00:00:00.144 |

Low-rated listings represent a direct risk to guest satisfaction, repeat bookings, and platform reputation. These properties are likely experiencing issues related to cleanliness, communication, amenities, or expectation mismatch. Airbnb should create an automated intervention program that flags listings below 4.0 and requires corrective actions from hosts. Performance should be tracked by reducing the number of sub-4.0 listings by at least 20% within the next review cycle.

## Host Engagement

Hosts are the backbone of Airbnb’s platform, and their success directly impacts the company’s growth. Hosts managing multiple listings contribute significantly to the platform’s inventory and revenue. Airbnb wants to identify its most engaged hosts and understand their performance.

1. Identify hosts managing more than 3 listings.



```
SELECT  
  
    host_id, COUNT(id) AS number_of_listings  
  
FROM listings  
  
GROUP BY host_id  
  
HAVING COUNT(id) > 3  
  
ORDER BY number_of_listings DESC;
```

|   | host_id<br>bigint | number_of_listings<br>bigint |
|---|-------------------|------------------------------|
| 1 | 232967878         | 25                           |
| 2 | 47625981          | 16                           |
| 3 | 526358730         | 11                           |
| 4 | 382970529         | 10                           |
| 5 | 349460842         | 10                           |
| 6 | 152762198         | 9                            |

Hosts managing multiple listings contribute disproportionately to platform supply and likely represent professional operators rather than casual hosts. Their retention is strategically important because losing one such host can significantly reduce inventory. Airbnb should introduce a dedicated account management and support program for high-volume hosts and monitor host retention rates and listing growth within this segment.

2. Calculate the average review score for each host across their listings.



```
SELECT  
  
    host_id, id,  
  
    ROUND(AVG(review_scores_rating)::NUMERIC, 2) AS avg_review_score  
  
FROM listings  
  
GROUP BY host_id, id  
  
ORDER BY avg_review_score;
```



|                 | host_id<br>bigint | id<br>bigint                | avg_review_score<br>numeric |
|-----------------|-------------------|-----------------------------|-----------------------------|
| 1               | 382446799         | 1190000000000000000         | 3.00                        |
| 2               | 433884293         | 5590000000000000000         | 3.00                        |
| 3               | 121680792         | 42389085                    | 3.00                        |
| 4               | 461579260         | 6870000000000000000         | 3.25                        |
| 5               | 461579260         | 6820000000000000000         | 3.33                        |
| 6               | 188679144         | 8430000000000000000         | 3.50                        |
| Total rows: 384 |                   | Query complete 00:00:00.115 |                             |

Host-level ratings provide a clearer measure of operational quality than individual listing ratings. Hosts consistently maintaining high ratings demonstrate scalable service quality across multiple properties. Airbnb should identify top-performing hosts and establish a mentorship framework where their practices are shared with newer hosts. Success should be measured through improvements in average host ratings among newly onboarded hosts.

3. List hosts with at least 2 listings and an average review score below 4.0.

|                                                                                     |                                                               |
|-------------------------------------------------------------------------------------|---------------------------------------------------------------|
|  | SELECT                                                        |
|                                                                                     | host_id, COUNT(host_id) AS total_listing,                     |
|                                                                                     | ROUND(AVG(review_scores_rating), 2) AS avg_rating_score       |
|                                                                                     | FROM listings                                                 |
|                                                                                     | GROUP BY host_id                                              |
|                                                                                     | HAVING COUNT(host_id) > 2 AND AVG(review_scores_rating) > 4.0 |
|                                                                                     | ORDER BY avg_rating_score DESC;                               |

|                | host_id<br>bigint | total_listing<br>bigint     | avg_rating_score<br>numeric |
|----------------|-------------------|-----------------------------|-----------------------------|
| 1              | 36438637          | 4                           | 4.65                        |
| 2              | 47625981          | 16                          | 4.65                        |
| 3              | 116935236         | 4                           | 4.91                        |
| 4              | 49100926          | 4                           | 4.54                        |
| 5              | 135717458         | 3                           | 4.82                        |
| 6              | 232967878         | 25                          | 4.79                        |
| Total rows: 39 |                   | Query complete 00:00:00.107 |                             |

Hosts successfully managing multiple listings while maintaining strong guest ratings demonstrate an ability to scale operations without compromising customer experience. Airbnb should prioritize these hosts for Superhost recognition, promotional visibility, and beta-testing new host tools. Tracking booking growth and retention within this host segment will help measure the impact of these initiatives.

## Booking Trends

Occupancy rates are one of the most critical indicators of a listing's success. Listings with high occupancy rates indicate strong demand, while low-occupancy listings may signal pricing issues or uncompetitive offerings.

Airbnb's sales team is keen to understand which listings perform well and which are underutilised during specific periods, such as January 2023. This insight will help them optimise marketing strategies and identify patterns in guest booking behaviour.

1. Calculate the occupancy rate for each listing for January 2023.



```
SELECT
    id, host_since_date,
    ((host_listings_count) / (host_total_listings_count)) * 100 AS occupancy_rate
FROM listings

WHERE host_since_date BETWEEN '2023-01-01' AND '2023-01-31';
```

|               | id<br>bigint | host_since_date<br>date     | occupancy_rate<br>integer |
|---------------|--------------|-----------------------------|---------------------------|
| 1             | 934000       | 2023-01-02                  | 100                       |
| 2             | 804000       | 2023-01-02                  | 100                       |
| 3             | 804000       | 2023-01-02                  | 100                       |
| 4             | 796000       | 2023-01-02                  | 100                       |
| 5             | 796000       | 2023-01-02                  | 100                       |
| 6             | 802000       | 2023-01-02                  | 100                       |
| 7             | 102000       | 2023-01-19                  | 100                       |
| Total rows: 7 |              | Query complete 00:00:00.121 |                           |

Occupancy rates reveal how effectively listings convert availability into bookings. High occupancy indicates strong demand, competitive pricing, and attractive property offerings, while low occupancy signals revenue leakage. Airbnb should identify common factors among high-performing listings and

incorporate those factors into listing optimization recommendations. A measurable goal would be improving occupancy rates among bottom-quartile listings by 5–10%.

2. Identify the top 5 listings with the highest occupancy rates in January 2023.



```
SELECT
    id, host_since_date,
    ((host_listings_count) / (host_total_listings_count)) * 100 AS occupancy_rate
FROM listings
WHERE host_since_date BETWEEN '2023-01-01' AND '2023-01-31'
ORDER BY occupancy_rate DESC
LIMIT 5;
```

|               | id<br>bigint | host_since_date<br>date     | occupancy_rate<br>integer |
|---------------|--------------|-----------------------------|---------------------------|
| 1             | 934000       | 2023-01-02                  | 100                       |
| 2             | 804000       | 2023-01-02                  | 100                       |
| 3             | 804000       | 2023-01-02                  | 100                       |
| 4             | 796000       | 2023-01-02                  | 100                       |
| 5             | 796000       | 2023-01-02                  | 100                       |
| Total rows: 5 |              | Query complete 00:00:00.114 |                           |

The highest-occupancy listings demonstrate strong product-market fit and effective pricing strategies. Understanding what differentiates these properties can help Airbnb improve performance across the broader marketplace. The host success team should conduct comparative analysis of amenities, pricing, and location attributes and use the findings to create optimization guidelines for underperforming listings.

3. List all listings that were not booked at all in January 2023.

|                                                                                   |                                                                |
|-----------------------------------------------------------------------------------|----------------------------------------------------------------|
|  | SELECT                                                         |
|                                                                                   | listing_id                                                     |
|                                                                                   | FROM calender                                                  |
|                                                                                   | WHERE date2 BETWEEN '2024-01-01' AND '2024-01-31'              |
|                                                                                   | GROUP BY listing_id                                            |
|                                                                                   | HAVING SUM( CASE WHEN available = 't' THEN 1 ELSE 0 END ) = 0; |

Listings with no bookings represent unrealized inventory and lost revenue opportunities. These listings may suffer from poor visibility, weak reviews, uncompetitive pricing, or incomplete profiles. Airbnb should launch an automated diagnostic program that identifies the primary reason for underperformance and recommends corrective actions. Success should be measured through reductions in inactive listings and improvements in first-booking conversion rates.

## Pricing Patterns Across Property Types

Airbnb’s pricing strategy team wants to understand how property type influences nightly rates. Identifying undervalued property types can guide pricing adjustments, while understanding high-value categories can inform marketing priorities.

Analysing pricing patterns will also help identify trends in demand and inform strategic decisions about property offerings.

1. Calculate the average price per night for each property type.

|                                                                                     |                                                          |
|-------------------------------------------------------------------------------------|----------------------------------------------------------|
|  | SELECT                                                   |
|                                                                                     | listing_id,                                              |
|                                                                                     | ROUND(AVG(price)::NUMERIC, 2) AS average_price_per_night |
|                                                                                     | FROM calender                                            |
|                                                                                     | GROUP BY listing_id;                                     |

|   | listing_id<br>bigint | average_price_per_night<br>numeric |
|---|----------------------|------------------------------------|
| 1 | 437362               | 179.00                             |
| 2 | 881000               | 89.00                              |
| 3 | 662333               | 125.00                             |
| 4 | 807000               | 100.00                             |
| 5 | 849000               | 270.00                             |

Average nightly pricing provides insight into market positioning and affordability. Significant pricing variation suggests the existence of multiple customer segments with different willingness to pay. Airbnb should use these findings to strengthen its dynamic pricing engine and improve price recommendations. Performance should be measured through increases in revenue per available night while maintaining occupancy levels.

2. Identify the top 5 listings with the highest average price per night and their property type.



```

SELECT
    C.listing_id,
    ROUND(AVG(C.price)::NUMERIC, 2) AS avg_price_per_night,
    L.property_type
FROM calender AS C
JOIN listings AS L
    ON C.listing_id = L.id
GROUP BY C.listing_id, L.property_type
ORDER BY avg_price_per_night DESC LIMIT 5;

```

|               | listing_id<br>bigint | avg_price_per_night<br>numeric | property_type<br>text |
|---------------|----------------------|--------------------------------|-----------------------|
| 1             | 929000               | 4000.00                        | Entire rental unit    |
| 2             | 622000               | 1350.00                        | Entire home           |
| 3             | 117000               | 874.50                         | Entire rental unit    |
| 4             | 117000               | 874.50                         | Entire home           |
| 5             | 117000               | 874.50                         | Private room in ho... |
| Total rows: 5 |                      | Query complete 00:00:00.444    |                       |

Premium-priced listings target high-value travelers and contribute disproportionately to revenue despite lower booking volumes. These properties require exceptional guest experiences to justify their pricing. Airbnb should develop luxury-focused marketing campaigns and monitor booking conversion rates within the premium segment to ensure pricing remains aligned with customer expectations.

3. Find property types with an average price below \$150 per night.



```
SELECT
    L.property_type,
    ROUND(AVG(C.price)::NUMERIC, 2) AS avg_price_per_night
FROM calender AS C
JOIN listings AS L
    ON C.listing_id = L.id
GROUP BY L.property_type
HAVING AVG(C.price) < 150.0;
```

|                | property_type<br>text     | avg_price_per_night<br>numeric |
|----------------|---------------------------|--------------------------------|
| 1              | Entire condo              | 120.86                         |
| 2              | Entire guest suite        | 91.00                          |
| 3              | Entire loft               | 115.33                         |
| 4              | Entire serviced apartm... | 118.50                         |
| 5              | Entire townhouse          | 131.92                         |
| Total rows: 13 |                           | Query complete 00:00:00.269    |

Property types averaging below \$150 per night appeal primarily to budget-conscious travelers and represent an important source of booking volume. Airbnb should highlight these categories in affordability-focused campaigns and introduce dedicated budget traveler filters. Success should be measured through increased booking volumes and improved occupancy within these property segments.

Guest Review Insights

---

Guest reviews offer invaluable insights into the strengths and weaknesses of Airbnb’s platform. Identifying frequent reviewers can help the company understand customer behaviour, while analysing listings with no reviews reveals areas of potential improvement in guest engagement.

1. Identify the top 10 reviewers by the total number of reviews submitted.



```
SELECT
    reviewer_id,
    COUNT(*) AS total_reviews
FROM reviews
GROUP BY reviewer_id
ORDER BY total_reviews DESC
LIMIT 10;
```

|   | reviewer_id<br>bigint | count<br>bigint |
|---|-----------------------|-----------------|
| 1 | 414682721             | 93              |
| 2 | 129172443             | 53              |
| 3 | 2450995               | 29              |
| 4 | 17224232              | 29              |
| 5 | 5373770               | 27              |

Frequent reviewers are highly engaged customers who provide valuable feedback and contribute significantly to platform trust. Airbnb should establish a reviewer recognition program that rewards consistent review participation. Success should be measured through increases in review submission rates and overall review coverage across completed stays.

2. Calculate the average number of reviews per listing.

|                                                                                   |                                                       |
|-----------------------------------------------------------------------------------|-------------------------------------------------------|
|  | SELECT                                                |
|                                                                                   | id,                                                   |
|                                                                                   | ROUND(AVG(number_of_reviews), 2) AS avg_no_of_reviews |
|                                                                                   | FROM listings                                         |
|                                                                                   | GROUP BY id;                                          |

|   | id<br>bigint | avg_no_of_reviews<br>numeric |
|---|--------------|------------------------------|
| 1 | 437362       | 123.00                       |
| 2 | 881000       | 11.00                        |
| 3 | 662333       | 328.00                       |
| 4 | 807000       | 32.00                        |
| 5 | 849000       | 1.00                         |

Listings with higher review volumes generally enjoy stronger customer trust and better booking conversion. Airbnb should focus on increasing review generation through automated reminders and post-stay engagement strategies. A measurable target would be increasing average reviews per listing by 15% annually.

3. Identify all listings with no reviews in 2023.

|                                                                                     |                                                             |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------|
|  | SELECT                                                      |
|                                                                                     | id                                                          |
|                                                                                     | FROM listings                                               |
|                                                                                     | WHERE host_since_date BETWEEN '2023-01-01' AND '2023-12-31' |
|                                                                                     | GROUP BY id                                                 |
|                                                                                     | HAVING SUM(number_of_reviews) = 0;                          |

Listings without reviews face a credibility disadvantage because potential guests lack social proof. Airbnb should prioritize these listings in review-generation campaigns and encourage hosts to provide exceptional first-stay experiences. Success should be measured by reducing the percentage of review-less listings and increasing first-review acquisition rates.



## Customer Engagement Trends

The team wants to analyze customer engagement across all listings within a specific time period. By identifying and ranking reviews submitted between two dates, the business can understand trends in customer interactions and review activity during that timeframe.

Rank all reviews submitted between '2015-01-01' and '2015-06-30' by their submission date.

|                                                                                   |                                                               |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------|
|  | SELECT                                                        |
|                                                                                   | listing_id,                                                   |
|                                                                                   | date2 AS review_date,                                         |
|                                                                                   | reviewer_name,                                                |
|                                                                                   | DENSE_RANK() OVER ( ORDER BY date2) AS rank_of_review_by_date |
|                                                                                   | FROM reviews                                                  |
|                                                                                   | WHERE date2 BETWEEN '2015-01-01' AND '2015-06-30';            |

|                | listing_id<br>bigint | review_date<br>date         | reviewer_name<br>text | rank_of_review_by_date<br>bigint |
|----------------|----------------------|-----------------------------|-----------------------|----------------------------------|
| 1              | 382021               | 2015-01-01                  | Molly                 | 1                                |
| 2              | 299245               | 2015-03-04                  | Altay                 | 2                                |
| 3              | 299245               | 2015-03-25                  | John                  | 3                                |
| 4              | 382021               | 2015-04-16                  | Macy                  | 4                                |
| 5              | 382021               | 2015-04-18                  | Ben                   | 5                                |
| Total rows: 37 |                      | Query complete 00:00:00.110 |                       |                                  |

Review activity trends provide insight into periods of peak customer engagement and booking activity. Higher review volumes during specific periods indicate seasonal demand patterns and marketing effectiveness. Airbnb should align promotional campaigns with historically high-engagement periods and monitor review growth as a leading indicator of marketplace activity.

## Most Active Hosts

The management team wants to identify the most active hosts based on the total number of reviews their listings have received. This analysis will help the team recognize high-performing hosts who maintain popular and engaging properties. The total number of reviews for each host is calculated by summing up the reviews of all their listings.

Calculate the total number of reviews received by each host across all their listings. Rank the hosts based on the total number of reviews.



```
SELECT

    host_id,

    host_name,

    SUM(number_of_reviews) AS total_number_of_reviews,

    DENSE_RANK() OVER ( ORDER BY SUM(number_of_reviews) DESC ) AS rank_of_host

FROM listings

GROUP BY host_id, id, host_name;
```

|                 | host_id<br>bigint | host_name<br>text           | total_number_of_reviews<br>bigint | rank_of_host<br>bigint |
|-----------------|-------------------|-----------------------------|-----------------------------------|------------------------|
| 1               | 9666970           | Janet                       | 853                               | 1                      |
| 2               | 5691268           | Elizabeth                   | 697                               | 2                      |
| 3               | 44434571          | Earl                        | 648                               | 3                      |
| 4               | 17766924          | Sugey                       | 598                               | 4                      |
| 5               | 44434571          | Earl                        | 574                               | 5                      |
| Total rows: 384 |                   | Query complete 00:00:00.172 |                                   |                        |

Hosts receiving the highest volume of reviews are likely generating the most guest interactions and maintaining strong booking activity. These hosts are critical contributors to platform engagement and revenue generation. Airbnb should focus retention efforts on this segment and track host churn rates, review growth, and booking performance to ensure continued marketplace stability.

Revenue for Top Listings

---

The finance team wants to identify the top revenue-generating listings. By analyzing the total revenue earned by each listing, the team can prioritize these listings for promotional strategies or investment. The revenue is calculated by summing up the price for all available days in the calendars table.

Calculate the total revenue for each listing by summing up the price for all available days. Use a window function to rank the listings based on their total revenue in descending order.



```
SELECT  
  
    listing_id,  
  
    SUM((price) * (CASE WHEN available ='t' THEN 1 else 0 end)) AS Total_revenue,  
  
    DENSE_RANK () OVER (ORDER BY SUM((price) * (CASE WHEN available ='t' THEN 1 ELSE 0  
END)) DESC) AS Rank_of_list_by_revenue  
  
FROM calender  
  
GROUP BY listing_id
```

|                 | listing_id<br>bigint | total_revenue<br>numeric    | rank_of_list_by_revenue<br>bigint |
|-----------------|----------------------|-----------------------------|-----------------------------------|
| 1               | 117000               | 1109036.00                  | 1                                 |
| 2               | 929000               | 840000.00                   | 2                                 |
| 3               | 938000               | 575546.00                   | 3                                 |
| 4               | 622000               | 448200.00                   | 4                                 |
| 5               | 101000               | 324861.00                   | 5                                 |
| Total rows: 301 |                      | Query complete 00:00:00.136 |                                   |

Revenue concentration among a small number of listings indicates that specific property characteristics, locations, or host practices significantly outperform the marketplace average. Airbnb should identify the key drivers behind these listings and build data-driven recommendations for other hosts. Success should be measured through revenue growth among mid-tier listings adopting similar practices.

## Property Type Performance in Summer

The marketing team wants to identify how property types perform relative to each other during the summer months. By calculating percentile ranks for property types based on their total summer revenue, the team can classify property types into performance tiers, such as top-80 percentile performing property types.

Calculate the total summer revenue (June, July, and August) for each property type. Determine its percentile rank among all property types based on their revenue.

|                                                                                   |                                                                                                                       |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
|  | SELECT                                                                                                                |
|                                                                                   | L.property_type,                                                                                                      |
|                                                                                   | SUM((C.price)*(CASE WHEN C.available ='t' THEN 1 ELSE 0 END)) AS Total_revenue,                                       |
|                                                                                   | PERCENT_RANK () OVER(ORDER BY SUM((C.price)*(CASE WHEN C.available ='t' THEN 1 ELSE 0 END ))) *100 AS Percentile_Rank |
|                                                                                   | FROM listings AS L                                                                                                    |
|                                                                                   | JOIN calender AS C                                                                                                    |
|                                                                                   | ON L.id = C.listing_id                                                                                                |
|                                                                                   | WHERE EXTRACT(MONTH FROM date2) IN (6, 7, 8)                                                                          |
|                                                                                   | GROUP BY L.property_type                                                                                              |
|                                                                                   |                                                                                                                       |

|                 | listing_id<br>bigint | total_revenue<br>numeric    | rank_of_list_by_revenue<br>bigint |
|-----------------|----------------------|-----------------------------|-----------------------------------|
| 1               | 117000               | 1109036.00                  | 1                                 |
| 2               | 929000               | 840000.00                   | 2                                 |
| 3               | 938000               | 575546.00                   | 3                                 |
| 4               | 622000               | 448200.00                   | 4                                 |
| 5               | 101000               | 324861.00                   | 5                                 |
| Total rows: 301 |                      | Query complete 00:00:00.191 |                                   |

Certain property types generate substantially higher revenue during summer months, reflecting strong seasonal demand patterns. Airbnb should increase promotional investments for these property categories before peak travel periods and encourage hosts to optimize pricing during high-demand seasons. Performance should be measured through seasonal revenue growth and booking share gains within targeted property types.



## Year-over-Year (YoY) Growth in Listing Revenue

The finance team wants to analyze how listing revenue changes over time by comparing year-over-year (YoY) growth. This information can help identify trends in revenue generation and highlight listings that are growing or declining in performance.

For each listing, calculate the total revenue per year and determine its YoY growth as a percentage. Use window functions to retrieve the previous year's revenue for each listing and calculate the YoY growth. Finally, rank listings based on their YoY growth percentage to identify top-performing listings.



```
WITH my_cte AS (  
  
    SELECT  
  
        listing_id,  
  
        EXTRACT(YEAR FROM date2) AS year,  
  
        SUM((price) * (CASE WHEN available = 't' THEN 1 ELSE 0 END)) AS Total_revenue  
  
    FROM calender  
  
    GROUP BY year, listing_id  
  
    ORDER BY listing_id, year)  
  
SELECT *,  
  
    LAG(total_revenue,1) OVER(PARTITION BY listing_id) AS pre_value,  
  
    ROUND((Total_revenue - (LAG(total_revenue,1) OVER(PARTITION BY  
listing_id))) * 100 / NULLIF(LAG(total_revenue,1) OVER(PARTITION BY listing_id),0),2) AS YOY_growth  
  
FROM my_cte
```

|                 | listing_id<br>bigint | year<br>numeric             | total_revenue<br>numeric | pre_value<br>numeric | yoy_growth<br>numeric |
|-----------------|----------------------|-----------------------------|--------------------------|----------------------|-----------------------|
| 1               | 299245               | 2023                        | 0.00                     | [null]               | [null]                |
| 2               | 299245               | 2024                        | 11060.00                 | 0.00                 | [null]                |
| 3               | 382021               | 2023                        | 11000.00                 | [null]               | [null]                |
| 4               | 382021               | 2024                        | 31000.00                 | 11000.00             | 181.82                |
| 5               | 565157               | 2023                        | 2550.00                  | [null]               | [null]                |
| Total rows: 602 |                      | Query complete 00:00:00.230 |                          |                      |                       |

Listings demonstrating strong YoY revenue growth indicate successful pricing, marketing, or operational strategies, while declining listings signal potential competitive or quality concerns. Airbnb should use high-growth listings as benchmark cases and proactively investigate declining properties. Success should be measured by increasing the proportion of listings achieving positive YoY revenue growth and reducing the number experiencing revenue decline.